

Conversational Constrained Clustering (C3)

1. Introduction

Text clustering is a key technique for organizing large amounts of unstructured data. However, unlike classification where labels are fixed, clustering is highly subjective. What makes a clustering result "correct" depends entirely on what the user is trying to find [Campello et al., 2020]. For example, a dataset of e-commerce reviews can be grouped by product features, customer feelings, or user types. Because there is no single perfect answer, researchers introduced **Interactive Clustering** [Balcan and Blum, 2008; Bae et al., 2020]. By bringing humans into the process, interactive systems allow users to guide classical algorithms, like K -Means [MacQueen, 1967] or Agglomerative Clustering [Ward, 1963], toward the results they actually want.

A highly effective way to guide these algorithms is **Constrained Clustering** [Wagstaff et al., 2001; Gançarski et al., 2020]. This method uses background knowledge in the form of simple rules—usually *Must-link* (two items must be in the same group) and *Cannot-link* (two items must be in different groups). These rules have been successfully built into algorithms like PCK-Means [Bilenko et al., 2004] and Constrained Spectral Clustering [Wang and Davidson, 2010]. However, traditional constrained clustering requires users to provide all these rules beforehand. This creates a difficult problem: users are asked to link specific data points before they even have a chance to explore or understand the data.

To solve this "cold-start" problem, **Interactive Constrained Clustering (ICC)** [Basu et al., 2004] was developed. In ICC, the system shows an initial clustering result, and users can add Must-link or Cannot-link rules step-by-step to fix any mistakes they see. While this helps, it creates a new issue: a heavy mental burden and a "semantic gap". Users naturally think in broad concepts (e.g., "*separate shipping complaints from product flaws*"). However, the system forces them to speak in low-level math by manually searching for and selecting specific pairs of data points. This tedious manual work makes ICC very hard for regular users to adopt.

Recent progress in Large Language Models (LLMs) like GPT-4 offers a great way to bridge this gap, as they can easily understand natural language and human intent [OpenAI, 2023]. A straightforward approach is to use LLMs directly to group the data (*LLM-as-a-Clusterer*) [Viswanathan et al., 2023; Zhang et al., 2024]. However, this

hits a major computational bottleneck. Because the attention mechanism in LLMs has a quadratic complexity $O(N^2)$ [Vaswani et al., 2017], feeding thousands of data points into a prompt leads to high token costs, very slow response times, and a drop in reasoning quality known as long-context hallucination [Liu et al., 2024]. As a result, direct LLM clustering remains highly unscalable.

To get the best of both worlds—the language understanding of LLMs and the speed of traditional algorithms—we propose **Conversational Constrained Clustering (C3)**. In our approach, the LLM does not cluster the data directly. Instead, it acts as a smart middle layer. It chats with the user to understand their high-level goals. Once the LLM understands what the user wants, it automatically translates those chat instructions into Must-link and Cannot-link rules on a small sample of the data. These rules are then sent to a fast, traditional algorithm (like PCK-Means) to process the entire dataset. With the C3 framework, users can easily guide massive datasets just by chatting, without losing the efficiency of classical machine learning algorithms.

2. Related Work

Our work lies at the intersection of interactive constrained clustering, large language models for data exploration, and conversational machine learning alignment. In this section, we review the state-of-the-art (SOTA) developments in these domains and position our Conversational Constrained Clustering (C3) framework within the literature.

2.1 Interactive and Constrained Clustering

Constrained clustering has a rich history of integrating user-defined prior knowledge into unsupervised algorithms to guide partitioning. Traditional algorithms rely on instance-level constraints, specifically Must-link and Cannot-link relations, to modify either the objective function or the distance metric. Classical approaches such as COP-KMeans [Wagstaff et al., 2001] enforce these constraints strictly during assignment, while soft-constrained algorithms like PCK-Means [Bilenko et al., 2004] and Constrained Spectral Clustering [Wang and Davidson, 2010] balance spatial data density against constraint violations using penalty metrics.

To make constraint collection feasible, **Interactive Constrained Clustering (ICC)** incorporates active learning strategies to query the user for constraints on-the-fly [Basu et al., 2004; Settles, 2012]. Advanced interactive systems use uncertainty sampling or expected utility maximization to select the most informative pairwise queries [Xiong et al., 2014; Yi et al., 2020].

The extensive research in this domain has been organized into several systematic surveys and frameworks. Gançarski et al. [2020] mapped historical trends in constrained clustering, while Kuncheva et al. [2022] provided a comprehensive bibliographic view tracking the distribution of research across disciplines. More recently, González-Almagro et al. [2025] introduced a detailed taxonomy of constraint types and applications, and Dao and Vrain [2024] synthesized declarative approaches that formulate constraints using exact constraint programming or propositional logic. In terms of user interaction, scoping and comprehensive reviews by Bae et al. [2020] and Neubauer et al. [2021] evaluated the diverse interactive mechanisms designed to put "humans-in-the-loop," confirming that traditional systems remain fundamentally bottlenecked by the **semantic gap** and the high **cognitive load** of low-level point-to-point annotations. Our work directly addresses this bottleneck by replacing manual pairwise queries with an intuitive conversational interface, allowing users to direct the clustering process using natural language.

2.2 LLMs for Unsupervised Text Exploration

With the rapid advancement of Large Language Models (LLMs), recent research has explored using generative models to assist in unstructured data exploration and clustering. A common paradigm, often referred to as **LLM-as-a-Clusterer**, bypasses traditional machine learning backends by prompting an LLM to read a collection of texts and directly output the category partitions [Viswanathan et al., 2023; Zhang et al., 2024]. Other works utilize LLMs to dynamically generate human-readable labels for pre-computed clusters, bridging the interpretability gap of traditional unsupervised algorithms [Chandra et al., 2023].

Recent evaluations of LLM embeddings for document clustering show that while larger models (such as proprietary GPT architectures) provide high-quality semantic representations, there exists a trade-off regarding computational efficiency, dimensionality reduction, and sensitivity to class imbalances [Keraghel et al., 2024]. Furthermore, a comprehensive, bottom-up survey of the evolving research on Large Language Model limitations (**LLMs**) highlights reasoning failures, context length constraints, and hallucinations as persistent challenges that hinder direct deployment for long-text analytical tasks [Kostikova et al., 2026].

Our C3 framework breaks this scalability-accuracy trade-off. Instead of employing the LLM as the heavy execution engine for clustering, we deploy it as an intelligent middleware that translates user intents into symbolic constraints over a sampled subset. The actual scaling and partitioning are offloaded to highly optimized

traditional clustering algorithms operating at $O(N)$ complexity, ensuring both semantic intelligence and infinite scalability.

2.3 Conversational Machine Learning and Alignment

Our framework is also closely related to the emerging field of conversational machine learning, which aims to align AI systems with complex human preferences through natural dialogue [Ouyang et al., 2022]. Conversational alignment has been successfully applied to tasks such as interactive image editing [Laiti et al., 2026], program synthesis [Chen et al., 2021], and database querying [Zhong et al., 2017]. Conversational agents (CAs) have evolved rapidly from spoken dialogue systems to modern, LLM-based architectures [Gao et al., 2018; Kusal et al., 2024]. In task-oriented and information-seeking dialogue systems, generating synthetic multi-turn conversational data has emerged as a key research direction for model alignment [Soudani et al., 2026].

In the context of clustering, however, conversational systems are heavily under-explored. Existing conversational text analysis tools typically focus on search, summarization, or simple filtering, but rarely allow users to reshape the underlying cluster boundaries of a massive corpus dynamically. By introducing the C3 framework, we formalize the process of converting multi-turn conversational trajectories into spatial constraint matrices. This allows us to bring the power of state-of-the-art conversational alignment to unsupervised clustering, establishing a novel protocol where users can perform complex, multi-aspect data partitioning through a continuous feedback loop.

3. Task Definition and Problem Formulation

To formally define the Conversational Constrained Clustering (C3) task, we establish the mathematical representation of the inputs, the target output, and the essential constraints of the problem.

Let $X = \{x_1, x_2, \dots, x_N\}$ represent a dataset of N text documents. We assume that each document x_i can be mapped to a continuous vector space via an embedding function $\Phi(x_i) \in \mathbb{R}^d$, which captures the baseline semantic properties of the text. Rather than a static, one-time execution, C3 is an interactive process driven by a multi-turn conversation. At any given interaction turn t , the conversational state is represented by the trajectory history $H_t = \{(u_1, a_1), (u_2, a_2), \dots, (u_t, a_t)\}$, where u_i is the user's natural language instruction and a_i is the corresponding system feedback.

The core objective of Conversational Constrained Clustering is to find an optimal sequential partitioning function $g : (X, H_t, C_{t-1}) \rightarrow C_t$ that groups the dataset X into K disjoint clusters $C_t = \{C_{t,1}, C_{t,2}, \dots, C_{t,K}\}$, where C_{t-1} represents the partition configuration generated in the previous turn. Under this state-transition formulation, a mathematically valid partition C_t must satisfy two competing objectives. First, it must maximize **Semantic Cohesion**, ensuring that documents grouped together share high vector similarity under the embedding function Φ . Second, it must maximize **Intent Alignment**, meaning that the boundaries of the partition C_t must conform to the high-level semantic criteria and structural rules implicitly or explicitly specified by the user in the conversation history H_t , while maintaining structural and semantic transition consistency from the prior state C_{t-1} .

4. Research Questions, Scope, and Assumptions

To maintain academic rigor and ensure our evaluations are fair and reproducible, we establish the experimental targets and structural boundaries of the C3 research framework.

4.1 Research Questions

To systematically evaluate the performance and feasibility of our conversational paradigm, our experimental design is structured to answer three fundamental research questions:

- **RQ1 (Translation Accuracy):** How accurately can the Large Language Model parse complex, abstract, and potentially ambiguous natural language directives from the user and translate them into mathematically precise Must-link and Cannot-link constraints?
- **RQ2 (Scalability and Speed):** Does the decoupled architecture of C3 successfully maintain linear execution scalability $O(N)$ and low operational latency as the scale of the dataset N grows, compared to direct end-to-end LLM clustering baselines?
- **RQ3 (Conversational Convergence):** How many continuous interaction turns are required for C3 to converge stably to the user's latent target partitioning, and how resilient is this convergence to optimization instabilities or semantic drift?

4.2 Scope and Assumptions

To ground our evaluations, we establish two key operational assumptions and define the explicit scope of the system:

- **Assumption 1 (Consistent User Intent):** We assume that within a single interactive session, the user's exploration goal is logically progressive and self-consistent. While users may iteratively refine, narrow, or expand their categorization criteria, they will not introduce fundamentally contradictory rules without explicitly requesting a state reset.
- **Assumption 2 (Capable Text Embeddings):** We assume that the underlying text embedding model provides a reasonable representation of basic semantic similarity in its vector space, as supported by empirical studies on LLM embedding structures [Keraghel et al., 2024]. The primary task of C3 is not to repair fundamentally flawed or noisy vector representations, but rather to adjust and refine the clustering boundaries of those representations based on user instructions.
- **Scope Limits:** Our system's scope is placed entirely on English-language textual datasets (such as customer logs, product reviews, or academic paper abstracts) within a dataset scale of 1,000 to 50,000 items, where live human-in-the-loop interaction remains practical and real-time feedback is computationally achievable.

5. Experimental Setup

In this section, we present the experimental design used to evaluate our Conversational Constrained Clustering framework. We detail the alternative multi-aspect dataset construction, define our proposed interactive metrics alongside classical clustering baselines, and describe the comparison models used to benchmark our system.

5.1 Dataset Construction

To evaluate a conversational clustering system, we require a dataset where the same set of documents can be grouped in multiple logically distinct ways, reflecting different potential user intents. Standard single-label datasets are insufficient for this task because they assume only one correct clustering result.

To address this, we formalize our evaluation under the framework of **Alternative Clustering** (also referred to as **Multi-Aspect** or **Multi-Facet Clustering**) [Bae et al., 2010]. In this paradigm, a single dataset X contains multiple independent, orthogonal ground-truth partitions, each representing a different semantic "aspect"

of the data. During interaction, the user's conversational intent acts as a query to select which target aspect the system should converge toward.

We construct our benchmark using two contrasting text datasets representing formal academic and informal colloquial language:

1. **arXiv Dataset (Academic Text):** We compile a corpus of scientific paper abstracts and their corresponding metadata from the arXiv repository [Clement et al., 2019]. This dataset is exceptionally well-suited for alternative clustering because a single abstract can be partitioned under multiple orthogonal aspects:
 - (1) the **Subject Domain Aspect** (partitioned into distinct academic domains such as High Energy Physics, Machine Learning, or Astrophysics based on the official, peer-reviewed `categories` metadata), and
 - (2) the **Methodological Aspect** (partitioned into theoretical/mathematical versus experimental/applied papers, which requires the LLM to perform deep semantic reasoning over the `abstract` text to identify the underlying research approach).
2. **Amazon Reviews Dataset (Colloquial Text):** We select reviews from the "Electronics" category of the Amazon Reviews dataset [Ni et al., 2019]. This represents a contrasting domain of colloquial, informal text. We establish two ground-truth aspects for evaluation:
 - (1) the **Sentiment Aspect** (partitioned based on star ratings into positive, neutral, and negative groups), and
 - (2) the **Feature Aspect** (partitioned based on the primary product attributes discussed, such as battery life, display quality, or customer service).

During evaluation, these target ground-truth partitions are kept completely hidden from both the embedding model and the clustering backend; they are only used to measure how successfully the conversational loop guides the system toward the user's selected aspect.

5.2 Evaluation Metrics

To measure how well the clustering output aligns with the user's intent across multiple interaction turns, we utilize both classical clustering metrics and a newly proposed interaction metric.

We employ the **Adjusted Rand Index (ARI)** [Hubert and Arabie, 1985] and **Normalized Mutual Information (NMI)** [Strehl and Ghosh, 2002] to calculate the similarity between the current cluster assignments at turn t and the targeted hidden

ground-truth view. These metrics output a score between 0 and 1, where 1 indicates a perfect match with the user's intended grouping and 0 represents random partitioning.

However, comparing a conversational system with traditional interactive clustering baselines poses a unique evaluation challenge. Traditional systems rely on manual, point-wise clicks to link specific pairs of data points, whereas our conversational system allows users to express complex directives using natural language. To ensure a fair comparison, we introduce a unified human-effort metric called **Interaction Cost Units (ICU)**, inspired by human-computer interaction cost models [Card et al., 1983].

We define a single point-wise manual link (a click) as 1 ICU. For conversational inputs, we assign a fixed cost of λ ICU per turn (where λ represents the average cognitive and temporal cost of reading a summary and typing a short instruction). In our experiments, we set $\lambda = 10$ ICU.

Using this standardized cost model, we evaluate the overall efficiency of the system using a new metric: **Area Under the Alignment Curve (AUAC)**.

Let $ARI(i)$ represent the Adjusted Rand Index of the system after spending i Interaction Cost Units of human effort. Within a total human effort budget of B units, the AUAC is mathematically defined as:

$$AUAC = \frac{1}{B} \sum_{i=1}^B ARI(i)$$

A highly efficient conversational agent will achieve a high ARI score after only one or two brief chat turns (consuming very few ICUs), resulting in a steep curve and a high AUAC score. Conversely, a traditional interactive system that requires tedious, point-by-point manual adjustments will improve slowly over many clicks, resulting in a lower AUAC score within the same budget.

5.3 Comparison Baselines

To demonstrate the advantages of our proposed framework, we compare its performance against three main classes of baseline models.

The first class represents **Unconstrained Baselines**, which perform traditional clustering without any human guidance. We use standard K -Means clustering [MacQueen, 1967] and Agglomerative Hierarchical Clustering [Ward, 1963] operating

directly on static text embeddings. These baselines establish the starting performance of the system before any user interaction occurs.

The second class consists of **Interactive Constrained Baselines**, which represent the traditional way of incorporating user feedback. We implement PCK-Means [Bilenko et al., 2004] and Constrained Spectral Clustering [Wang and Davidson, 2010]. In these systems, instead of chatting, the user is simulated as an oracle that manually selects specific pairs of data points and labels them as Must-link or Cannot-link based on the target ground-truth view. These baselines show how well the backend algorithm performs when guided by slow, precise, manual human effort.

The third class comprises **Direct LLM Baselines**, representing the naïve application of modern generative models. We implement a zero-shot *LLM-as-a-Clusterer* pipeline [Viswanathan et al., 2023] using models like GPT-4, where the model is prompted with the entire dataset and instructed to output the final cluster groupings directly in a single step. This baseline is used to measure the execution speed, token costs, and scalability bottlenecks of direct LLM processing compared to our hybrid conversational pipeline.

6. References

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